Multilayer Perceptron Formulae

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$$Z = \sum_{i=1}^{n} (x_i \cdot w_i) + b_i$$

Z is the linear combination and is not activated. x_i is your input (either a vector or an array).
w_i is the weight for each input. b_i is the bias for each neuron node.

$$A = f(Z)$$

A is activated. $f(\cdot)$ is activation function.
Activation functions:

Tanh:

$$tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

ReLU:

$$f(z) = max(0, z)$$

Sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Softmax:

$$Softmax(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}}$$

Forward propagation

- Calculating Z (linear combination of inputs)
- Applying activation function $A = f(Z)$
- Repeating layer-by-layer until final prediction $\hat{y}$

In layer L:
Weight: $W^{(L)}$, Biases: $b^{(L)}$, Non-activated combination: $Z^{(L)}$,

Activated output: $A^{(L)}$, Activation function: $f^{(L)}(\cdot)$.
Input:

$$x^{(L_{in})}$$

Output:

$$Z^{(L_{out})} = A^{(L_{hidden})} \times w^{(L_{out})} + b^{(L_{out})}$$
$$\hat{y} = f^{(L_{out})}(Z^{(L_{out})})$$

$\hat{y}$ is model predict output. Loss function:

Mean Squared Error, MSE

$$L(y, \hat{y}) = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$

Binary Cross-Entropy (for sigmoid output):

$$L(y, \hat{y}) = -\frac{1}{m}\sum_{i=1}^{m}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

Categorical Cross-Entropy (for softmax output):

$$L(y, \hat{y}) = -\sum_{i=1}^{m}(y_i \times \log(\hat{y}_i))$$

Backward propagation
Output Gradient

$$\delta^{(L_{out})} = \frac{\partial L}{\partial A^{(L_{out})}} \times f^{(L_{out})}(Z^{(L_{out})})$$
$$\frac{\partial L}{\partial W^{(L_{out})}} = A^{(L_{hidden})T} \times \delta^{(L_{out})}$$
$$\frac{\partial L}{\partial b^{(L_{out})}} = sum(\delta^{(L_{out})}, axis = 0)$$

Hidden Gradient

$$\delta^{(L-1)} = \delta^{(L)} \times (W^{(L)})^T \times f^{(L-1)}(Z^{(L-1)})$$
$$\frac{\partial L}{\partial W^{(L-1)}} = A^{(L-1)T} \times \delta^{(L-1)}$$
$$\frac{\partial L}{\partial b^{(L-1)}} = sum(\delta^{(L-1)}, axis = 0)$$

Update weights and biases(Gradient Descent)

$$W^{(L)}_{new} = W^{(L)}_{old} - learning\_rate \times \frac{\partial L}{\partial W^{(L)}}$$
$$b^{(L)}_{new} = b^{(L)}_{old} - learning\_rate \times \frac{\partial L}{\partial b^{(L)}}$$